

Design Specification

3-Axis Balancing Cube

Hardware Revision 1.0

1 • ABOUT THE PROJECT

This document outlines the electrical and mechanical architecture for a 15cm³ self-balancing reaction wheel robot. The system utilizes Model Predictive Control (MPC) or Linear Quadratic Regulation (LQR) to handle the coupled dynamics of three orthogonal axes.

Key Design Desicions

- **Dimensions:** 15cm × 15cm × 15cm
- **Power Source:** 4S LiPo (16.8V Peak)
- **Compute:** ESP32-S3 (Dual Core)
- **Motor Drivers:** TI DRV8316

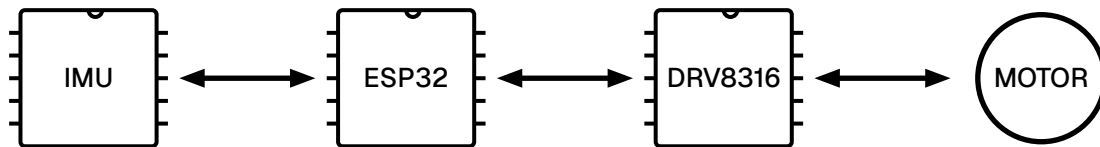


Figure 1: block diagram

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2 • SYSTEM ARCHITECTURE

The system is designed around a centralized controller (ESP32-S3) that manages three motor drivers via SPI and PWM. The control loop runs on a dedicated core to ensure real-time performance for the MPC solver.

2.1 • HIGH-LEVEL BLOCK DIAGRAM

- **Inputs:** IMU (SPI), Encoders (Quadrature), Battery Voltage (ADC).
- **Processing:** ESP32-S3-WROOM-1 Module.
- **Outputs:** 3x PWM signals (U, V, W) to Gate Drivers.
- **Actuation:** 3x BLDC Motors (Gimbal Type) via DRV8316.

3 • ELECTRONICS DESIGN

The PCB is a custom 4-layer board designed for high vibration resistance and thermal dissipation.

3.1 • MICROCONTROLLER SELECTION

Component: ESP32-S3-WROOM-1-N8R8

- **Form Factor:** Surface Mount Module (Castellated).
- **Reasoning:** Through-hole headers fail under reaction wheel vibration. The module provides robust mechanical coupling and optimal center of mass.
- **Peripherals Used:** - 3x PCNT (Hardware Encoder Counters)
 - 2x MCPWM (Motor Control PWM)
 - ADC1 (Current Sensing - ADC2 is avoided due to WiFi conflict)

3.2 • 3.2. PIN MAPPING (ESP32-S3)

Critical: Do not route current sensing to ADC2. Do not pull strapping pins (GPIO 0, 45, 46) high during boot.

Function	Signal	GPIO	Note
Motor X	PWM U/V/W	14, 15, 16	MCPWM Unit 0
	Current Sense	1	ADC1_CH0 (Strict requirement)
	Encoder A/B	38, 39	PCNT Unit 0
Motor Y	PWM U/V/W	17, 18, 21	MCPWM Unit 0
	Current Sense	2	ADC1_CH1
	Encoder A/B	40, 41	PCNT Unit 1
Motor Z	PWM U/V/W	35, 36, 37	MCPWM Unit 1
	Current Sense	3	ADC1_CH2
	Encoder A/B	42, 43	PCNT Unit 2
Shared	IMU SPI (SCK/MI/MO)	12, 13, 11	LSM6DSO Connection
	Driver nSLEEP	4	Active High to wake drivers
	Battery Sense	7	ADC1_CH6 (Voltage Divider)

Table 1: Master Pin Assignment Table

3.3 • 3.3. MOTOR DRIVERS (TI DRV8316)

Variant: DRV8316R (Hardware) or DRV8316T (SPI). SPI is preferred for adjustable gain.

- **Current Sensing:** The chip has internal Current Sense Amplifiers (CSA) but **no internal ADC**.

- **Wiring Requirement:** The S01, S02, S03 pins must be routed to the ESP32 ADC pins listed above.
- **Mode:** Operate in Current Sense Mode (VREF = 3.3V). Do not use the hardware “Current Limiting Mode” or the torque control will fight the MPC solver.

3.4 • 3.4. POWER REGULATION STRATEGY

Direct 16.8V to 3.3V conversion is thermally impossible for the ESP32 load.

1. Stage 1 (Buck): TPS54302 converts 16.8V → 5.0V.
2. Stage 2 (LDO): AMS1117-3.3 converts 5.0V → 3.3V (Clean logic power).

4 • 4. SAFETY & PROTECTION CIRCUITS

Reaction wheels generate massive regenerative currents when braking. Protection is mandatory.

4.1 • 4.1. REVERSE POLARITY PROTECTION

Component: P-Channel MOSFET (e.g., AO4407).

- **Circuit:** Gate to GND, Source to Battery+, Drain to System.
- **Why:** Low voltage drop (<0.1V) compared to diodes, prevents explosion if battery is plugged in backwards.

4.2 • 4.2. REGENERATIVE BRAKING CLAMP

Passive: 1000μF 35V Electrolytic Capacitor at the main power entry.

- **Active (Optional):** If voltage > 18V, a comparator triggers an N-FET to dump energy into a 10Ω, 50W resistor.

4.3 • 4.3. BATTERY MANAGEMENT (BMS)

Module: Generic 4S 20A BMS (HX-4S-A01).

- **Function:** Prevents cell under-voltage (<3.0V) and short circuits.
- **Placement:** Located between the LiPo battery and the Main PCB.

5 • 5. CONTROL THEORY (MPC IMPLEMENTATION)

5.1 • 5.1. STATE SPACE MODEL

The robot is modeled as a coupled system. The state vector x contains 9 elements:

$$x = \left[\theta_x, \theta_y, \theta_z, \dot{\theta}_x, \dot{\theta}_y, \dot{\theta}_z, \omega_{wx}, \omega_{wy}, \omega_{wz} \right]^T$$

Where:

- θ : Cube orientation angle (Euler or Quaternion error).
- $\dot{\theta}$: Angular velocity of the cube (Gyro data).
- ω_w : Angular velocity of the reaction wheels (Encoder data).

5.2 • 5.2. CONTROL LAW

We utilize an LQR approach for the balancing loop:

$$u(t) = -K \cdot x(t)$$

Where K is the optimal gain matrix derived from the system physics (Inertia Tensor) and cost matrices Q and R .

5.3 • 5.3. SOFTWARE SANDBOX (SAFETY LIMITS)

To prevent MPC instability (hallucinations) from destroying hardware:

- **Max Tilt:** If $|\theta| > 20^\circ$, `Disable_Motors()`.
- **Max Current:** Software clamp at 6A per motor.
- **Slew Rate:** Limit $\frac{d(\text{Torque})}{dt}$ to prevent mechanical jerking.

6 • 6. BILL OF MATERIALS (KEY COMPONENTS)

Component	Part Number	Notes
MCU Module	ESP32-S3-WROOM-1	N8R8 (8MB Flash/PSRAM) recommended.
Motor Driver	TI DRV8316	Qty: 3. Requires bottom thermal pad vias.
IMU	ST LSM6DSO	SPI Interface. Mount at PCB center.
Buck Converter	TPS54302	Steps 16V down to 5V.
Protection FET	AO4407	P-Channel MOSFET for reverse polarity.
Bulk Cap	1000 μ F 35V	Low ESR Electrolytic.
Battery	4S LiPo 650mAh	High C-rating (45C+).

7 • 7. NEXT STEPS

1. Simulation: Model the physics in Python/MATLAB to derive the $\bar{K}$ matrix.
2. Schematic Capture: Implement the pin map in KiCad/EasyEDA.
3. PCB Layout: Focus on thermal vias for DRV8316 and minimizing loop inductance for the capacitors.